Lesson 2.10 Guided Notes

Session 1 · Mean or Median?

I can choose the mean or the median to describe a data set based on its shape. · 6.SP.B.5d

Word / Spanish	Meaning — complete the blank	Example
outlier / valor atípico	A value _____ away from the others.	2, 3, 4, 32: 32
skewed / asimétrico	Most values sit on one side, with a long _____.	tail to the right
symmetric / simétrico	Both sides of the data look about the _____.	20, 22, 24, 26, 28
measure of center / medida de centro	One number that shows a _____ value.	mean or median

Key idea

An outlier pulls the _____ toward it, but barely moves the median.

Skewed data or an outlier: use the _____. Symmetric data: use the mean.

Worked example

Five friends saved $6, $7, $8, $9 and $50. Which center describes a typical friend?

The outlier is _____.

Median: the middle of 6, 7, 8, 9, 50 is _____.

Mean: 80 ÷ 5 = _____, more than four of the five savings.

Answer: Use the _____: about $8.

You try

1. Write the better measure of center: mean or median.
a. 70, 72, 74, 76, 78 _____
b. 7, 8, 9, 10, 65 _____

_____

2. Comic book prices are $4, $5, $6, $7 and $53. Find the mean and median. Which is better?

_____

Sum it up: I use the median instead of the mean when ___ because ___.

_____

Session 2 · Matching Center and Spread

I can match a measure of center and a measure of spread to the shape of a data set. · 6.SP.B.5d

Word / Spanish	Meaning — complete the blank	Example
symmetric / simétrico	Both sides of the data look about the _____.	20, 22, 24, 26, 28
skewed / asimétrico	Most values sit on one side, with a long _____.	20, 22, 24, 26, 150
mean and MAD / media y DMA	The pair for _____ data with no outlier.	center plus average distance
median and IQR / mediana y rango intercuartil	The pair for skewed data or data with an _____.	middle plus middle half

Key idea

MAD is built from the _____; IQR is built from the median. Never mix the pairs.

Look for an outlier or a long _____ before you choose a pair.

Worked example

Set C: 20, 22, 24, 26, 28. Set D: 20, 22, 24, 26, 150. Pick a pair for each.

Set C is _____: mean 24, MAD 12 ÷ 5 = 2.4.

Set D has an outlier, _____: median 24.

Answer: C: mean and MAD. D: _____ and IQR.

You try

1. Write the pair you would report.
a. 40, 42, 44, 46, 48 _____
b. 40, 42, 44, 46, 300 _____

_____

2. Seven values are 22, 24, 26, 28, 30, 32, 320. Find the median and the IQR.

_____

Sum it up: I report the median and IQR when ___ because ___.

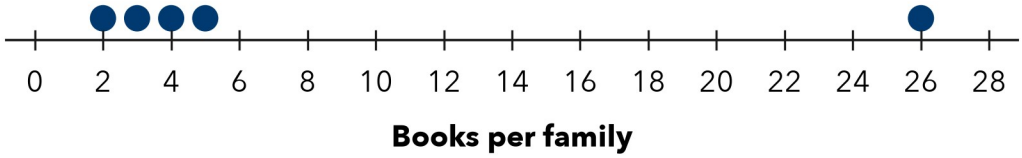
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Session 1 · Model, reason, and explain

Mean or Median? · 6.SP.B.5d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Five families collected 2, 3, 4, 5, and 26 books. Find the mean and median. Choose a measure for a typical family's collection, then for dividing the total equally among five delivery boxes.



_____

_____

_____

Extend your reasoning

Explain why calling the mean 'wrong' because of an extreme value is misleading.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Matching Center and Spread · 6.SP.B.5d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare A: 4,6,8,10,12 and B: 4,6,8,10,32. Describe shape, choose a center/spread pair, and justify your choice for describing typical observations. Use the stated median-of-halves convention.

A: 4, 6, 8, 10, 12

B: 4, 6, 8, 10, 32

_____

_____

_____

Extend your reasoning

How does the small sample affect a blanket claim that IQR ignores the largest value?

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?